Technical Datasheet — V11 Cell (t1_r3)

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Note: virtual design datasheet; every value mechanically taken from the parameter set / simulation outputs (`runs/exp/t1_r3/cell/r5_v11_*.json`). This is a simulation-caliber datasheet, not a production release.

Field	Value	Source
Rated capacity (Ah)	5.085 nominal; 5.0849 simulation-verified (1C, 25 °C)	`r5_v11_final.json` / `r5_v11_1c_spme.json:capacity_ah`
Nominal voltage / window (V)	4.130 V midpoint; 2.5 – 4.2 V window	`r5_v11_energy.json:midpoint_voltage_v` / Chen2020 cut-offs
Rated energy (Wh)	18.622	`r5_v11_energy.json:energy_wh`
Energy density (Wh/kg)	605.33	`r5_v11_energy.json:energy_density_wh_kg`
Volumetric energy density (Wh/L)	981.20	`r5_v11_energy.json:energy_density_wh_l`
Maximum continuous discharge rate	1C (5.085 A) verified by simulation; higher rates not simulated	`r5_v11_1c_spme.json`
Fast-charge capability	4C (20.34 A): 92.0% of capacity in 13.8 min, anode potential min +0.0425 V (no plating), T_max 321.2 K at 45 °C ambient	`r5_v11_4c.json`
Operating temperature range	Verified 25 °C (1C discharge, 298.15 K ambient) and 45 °C ambient (4C charge, 318.15 K ambient); beyond this range not simulated	protocol conditions (runner PROTOCOLS)
Cycle life	**Not simulated (requires aging model) — not fabricated**	—
Safety determination	4C plating: none (anode +0.0425 V); 4C T_max 321.2 K ≤ 333.15 K; overcharge 0.5C to 4.70 V: T_max 298.57 K, thermal-runaway triggered = false	`r5_v11_4c.json`, `r5_v11_oc.json`, `r5_v11_tr.json`
Dimensions and mass	single-layer stack: 65 mm × 1580 mm × 184.8 µm; 30.76 g (contract caliber, electrolyte excluded); 40.77 g incl. electrolyte; shell dims not parameterized	Chen2020 geometry + `r5_v11_energy.json`
Internal resistance (DCR)	19.25 µΩ (from 1C curve)	`r5_v11_energy.json:dcr_ohm`